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Highly imbalanced fault diagnosis of mechanical systems based on wavelet packet distortion and convolutional neural networks

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ABSTRACT

The healthy operations of mechanical systems are crucially important for ensuring human safety and economic benefits, so that there is a high demand on the automatic fault diagnosis techniques. However, the number of available faulty samples of mechanical systems is often far less than healthy samples, and thereby the traditional data-driven methods often suffer a high rate of misdiagnosis. In this paper, a new fault diagnosis method is developed on the basis of wavelet packet distortion and convolutional neural networks. First, wavelet packet distortion means that wavelet packet coefficients are distorted to augment fault samples, in order to achieve the equilibrium between healthy and faulty classes. Second, a convolutional neural network-based classification model is trained using the balanced training dataset. Third, the trained model is applied to classify the testing samples. Finally, the efficacy of this developed method in imbalanced fault diagnosis of mechanical systems is demonstrated through a number of experiments.

1. Introduction

There are many important mechanical systems in the industries. For example, aero hydraulic pumps are significant in supplying high pressure hydraulic oil for controlling the taking off, cruise, and landing of civil airplanes [1]. As a mechanical equipment with both rotating and reciprocating parts working under harsh environment, aero hydraulic pumps may suffer some unexpected faults. Therefore, accurate detection of incipient failures is significant for ensuring the flight safety [2,3].

The existing mechanical fault diagnosis technologies can be classified into three categories, including physics-based modelling, signal analysis-based methods, and data-driven methods [4]. To be specific, physics-based modelling builds the failure or degradation models according to physical rules, and predicts the behaviors based on the models. For example, Yang et al. [5] presented an improved mesh stiffness calculating model, in which the opened crack state is taken into consideration, so that the simulation accuracy was improved. Qin et al. [6] considered the influences of gyroscopic moment, centrifugal force, and time-varying contact angles, and established a novel high-speed fault dynamic model for angular contact ball bearings. However, the existing physics-based modelling methods mostly only focus on the simulation of very simple components, such as bearings and gears. The structures and dynamic behaviors of large machines are much more complex than these simple components. It is very challenging to build accurate physical models for large machines, especially under the real working conditions. The signal analysis-based fault diagnosis methods try to capture the failure characteristics from monitoring signals. For example, Teng et al. [7] applied Hilbert demodulation and cepstrum analysis to detect obvious gear faults, and applied complex wavelet transform to detect bearing cracks and looseness. Wang et al. [8] estimated instantaneous angular speeds without prior knowledge for fault detection of rolling bearings in civil aircraft engines. However, the monitoring data often contain much noise. Both the noise and failure characteristics change with the operating conditions. As a result, typical signal analysis methods often failed to detect failure characteristics.

In contrast, data-driven methods train automatic classification models using historical monitoring data, and apply the classification models for diagnosing the faults. Researchers do not need to reveal the physical principles of failures or degradations when using data-driven methods. In recent years, a variety of data-driven fault diagnosis methods have been developed [9]. For example, Du et al. [10] extracted a set of wavelet leaders multifractal features, and applied multi-class

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support vector machines for fault diagnosis. Chen et al. [11] decomposed vibration signals using wavelet analysis, extracted multi-domain features, and used optimized relevance vector machines with multiple kernels for fault diagnosis. Besides, it is worth mentioning that industrial processes can also be monitored using data-driven methods. For example, Márquez-Vera et al. [12] developed a very interesting inverse fuzzy fault model, in which least angle regression is used for variable selection, in order to detect and isolate faults. More than that, Márquez-Vera et al. [13] presented another very outstanding research, in which independent components analysis is used to reduce dimension, wavelet transform is used to highlight fault signals, and a long short term memory network is fed with a set of most representative variables selected by an elastic net. These research has effectively promoted the development of fault diagnosis techniques, and can be highly instructive to the future research. However, a potential problem in traditional data-driven methods is that their performance is highly dependent on the quality of manually crafted features. Once the features do not fully represent the characteristics of health states, these data-driven methods would completely fail to detect faults. Therefore, the construction of an informative set of features is crucially important to ensure the performance of data-driven fault diagnosis methods.

As an alternative to feature extraction, feature learning brings a new way to construct informative feature set [14,15]. The underlying idea is to allow or enforce the algorithm to automatically learn useful features from the raw monitoring signals, rather than configuring features by engineers [16]. The superiority of feature learning in improving diagnosing accuracy has been validated in some literatures [17–19]. For example, Cheng et al. [20] constructed a nonlinear degradation indicator using convolutional neural networks (ConvNets), and predicted the degradation indicator using a bidirectional long short-term memory network. Wang et al. [21] applied stacked autoencoders with batch normalizations (BNs) for automatic feature learning from monitoring signals, and yielded a high diagnosing accuracy. Huang et al. [22] used a residual gated dynamic sparse network to fuse and learn features from multi-sensor data, and validated the performance on a gearbox dataset. Shen et al. [23] used stacked contractive autoencoders for robust feature learning for fault diagnosis of rotating machines. In general, feature learning is achieved by using deep learning methods, *i.e.*, machine learning methods with multiple layers of nonlinear transformations.

However, traditional deep learning methods have a requirement that different classes have the same or similar numbers of training samples. If the dominant class has much more samples than the other classes, deep learning methods would tend to classify the majority of testing samples into the dominant class, which is generally known as the class-imbalanced issue [24]. Aiming at the problem, some research has been conducted. For example, Qin et al. [25] presented a dynamic over sampler with adaptive boosting for multi-class imbalanced classification problems. Tan et al. [26] employed a rebalanced mix-up training method together with domain adversarial training to generate a reliable distribution. Chen et al. [27] adopted random sampling to generate training samples for the minority classes, and achieved the balance between the classes. However, for the minority classes with extremely limited numbers of training samples, *e.g.*, 4 or 5 training samples, the existing methods cannot ensure a good performance. On one hand, some rebalanced methods with complex training processes are difficult to train. On the other hand, some simple random sampling methods cannot introduce enough diversity into the training dataset.

Aiming at the imbalanced issue, this paper develops a novel fault diagnosis approach on the basis of wavelet packet distortion and ConvNets. As the key step, wavelet packet distortion is developed for augmenting the fault samples. First, the vibration signals are decomposed into a number of wavelet coefficients using wavelet packet transform. Second, the wavelet coefficients at a stochastically selected wavelet packet node are randomly distorted. Third, the distorted wavelet coefficients are assembled together with the other wavelet coefficients, and used to reconstruct the samples. The procedure can be repeated, so that a large number of samples can be generated and serve as the training dataset. In the end, a ConvNet can be trained using the balanced dataset, which can achieve a relatively higher diagnosing accuracy for the minority class.

The rest of this paper is organized as follows. The theory and procedures of the developed imbalanced fault diagnosis method are elaborated in Section 2. The experimental results are summarized and discussed in Section 3. Finally, the conclusions are drawn in Section 4.

2. The developed imbalanced fault diagnosis approach

As mentioned in Section 1, this paper develops wavelet packet distortion on the basis of wavelet packet transform, in order to increase the number of faulty training samples. Here, the augmented samples are similar but have different values with the original samples. Thus, the balance among the classes can be achieved, and the sample diversity in the training dataset can be improved.

2.1. Wavelet packet distortion

As depicted in Fig. 1, the developed wavelet packet distortion method is mainly composed of three steps, including wavelet packet transform, distortion of a set of randomly selected wavelet coefficients, inverse wavelet packet transform. Wavelet packet transform [28] is a classical time–frequency analysis method that can decompose a signal to be a set of wavelet coefficients at different frequency bands, and has been widely applied in vibration-based fault diagnosis of mechanical systems [29–33]. On the contrary, inverse wavelet packet transform can be used to recover the signal using the wavelet coefficients; the reconstructed signal will be perfectly identical with the original sample if no other operation is performed on the wavelet coefficients. This research conducts distortion on a set of randomly selected wavelet coefficients, so that the reconstructed signal will be similar to but somewhat different from the original sample. The reconstructed signals can be used as the augmented samples, especially for the minority classes.

Let (h, g) be a pair of finite impulse response filters in length of L, and let $w_{j,i}$ denote the i th set of wavelet coefficients at the j th decomposition level. Specifically, $w_{0,0}$ is the original signal. According to the Mallat recursive algorithm, wavelet packet transform can be implemented recursively using the functions:

$$w_{j+1,2i}[k] = \sum_{l=0}^{L-1} w_{j,i}[2k-l]h[l] \tag{1}$$

$$w_{j+1,2i+1}[k] = \sum_{l=0}^{L-1} w_{j,i}[2k-l]g[l] \tag{2}$$

where $i = 0, \cdots, (2^j - 1)$, k and l are the sequential indexes for the elements. As depicted in Fig. 1, a two-level wavelet packet decomposition tree is conducted, so that there are 4 sets of wavelet coefficients.

Afterwards, one set of wavelet coefficients is stochastically selected, and nonlinearly distorted using the function:

$$\widetilde{w} = \mathrm{sign}(w) \bullet (\mathrm{abs}(\mathrm{w}))^{\mathrm{d}} \tag{3}$$

where d is a distortion coefficient that is stochastically selected from a preset range.

The distortion is the key step that makes the augmented sample different from the original sample. If d is close to 1, $\widetilde{w}$ will be close to w. In contrast, if d is far away from 1, there will be a large difference between $\widetilde{w}$ and w. Therefore, it is suggested to include 1 in the preset range. Considering that the fault characteristics are mostly associated with the frequency components, a benefit of the distortion operation is that the frequencies of the original sample can be mostly kept in the augmented samples, which makes the failure types of samples mostly unchanged. The effect of the distortion operation has been depicted in

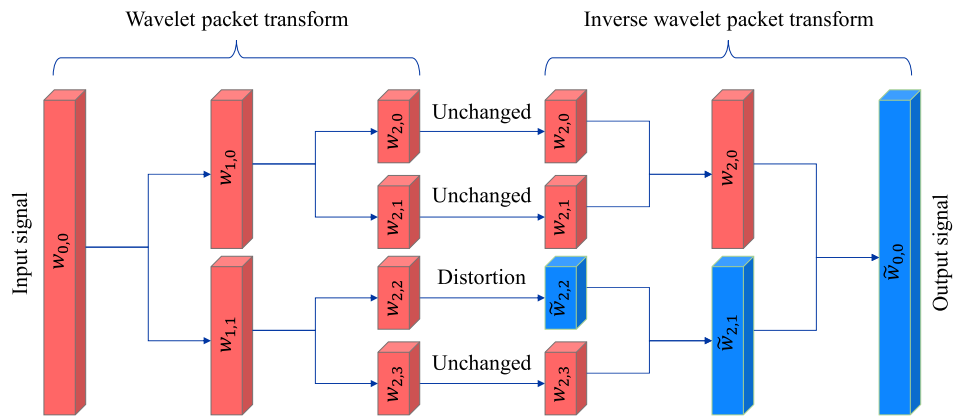
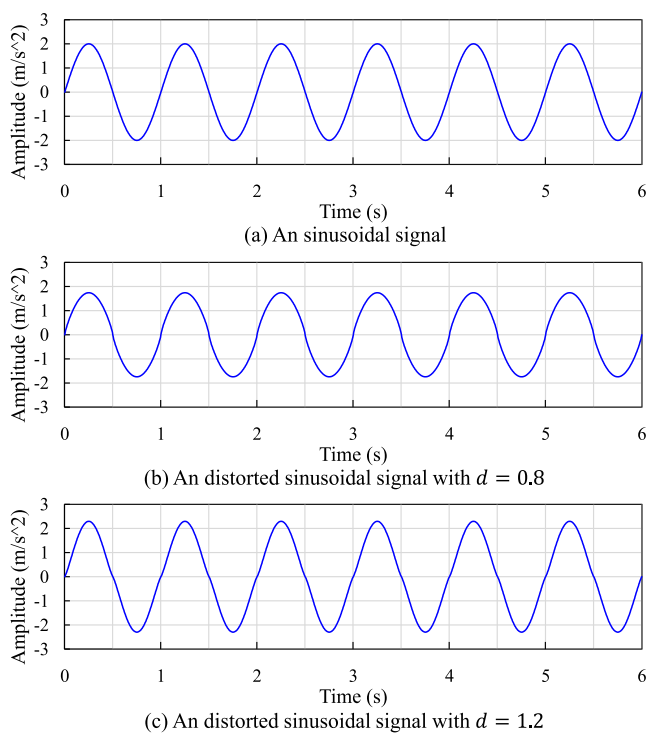


Fig. 1. The procedures of wavelet packet distortion.



(a) An sinusoidal signal

(b) An distorted sinusoidal signal with $d = 0.8$

(c) An distorted sinusoidal signal with $d = 1.2$

Fig. 2. An sinusoidal signal and its two distorted versions.

Fig. 2. The waveforms of distorted sinusoidal signals with $d = 0.8$ and $d = 1.2$ have similar shapes with the original sinusoidal signal.

After the distortion of some wavelet coefficients, inverse wavelet packet transform is conducted to obtain the output signal v, in which the formula is expressed by

$$v_{j,i}[k] = \sum_{l=0}^{L-1} \widetilde{w}_{j+1,2i}[(k-l)/2]\widetilde{h}[l] + \sum_{l=0}^{L-1} \widetilde{w}_{j+1,2i+1}[(k-l)/2]\widetilde{g}[l] \tag{4}$$

where $\widetilde{w}$ is the up-sampled-by-two version of w, and v is the reconstructed wavelet coefficients or signal. $(\widetilde{h}, \widetilde{g})$ are the pair of filters that is used in the reconstruction. Additionally, considering that the standard deviations of vibration signals are tightly related to the failures, this paper keeps the standard deviations of augmented samples the same with their corresponding original sample.

2.2. Fault feature learning via ConvNet

In recent years, ConvNet has become the most popular deep learning method in vibration -based fault diagnosis tasks [34–38]. The major superiority of ConvNet is that it can automatically learn informative features from raw monitoring signals, without the necessary of manually designing the features, so that no much specialized knowledge on signal processing and physical-of-failure mechanism is required.

The architecture of a typical ConvNet is depicted in Fig. 3, which is composed of a few convolutional layers (ConvLs), BNs, rectified linear units (ReLUs), global average pooling, fully connected layer, softmax, and cross-entropy loss. These components are described as below.

ConvL combines the concepts of local receptive fields and parameter sharing. First, the local receptive field means that a neuron is only linked to a small number of neurons in the previous layer. Second, the parameter sharing means that the weight of a neuron is shared with the other neurons at the same layer. Through the use of the two concepts, the number of trainable parameters can be greatly reduced, so that the deep ConvNets can be trained relatively easier than the general multi-layer perception networks. The formula of a ConvL can be expressed by

$$x_j^m = f\left(\sum_{i \in M_j} x_i^{m-1} * c_{ij}^m + b_j^m\right) \tag{5}$$

where x is the feature map, c is the convolution kernel, b is the bias, * indicates convolution, M_j is the collection of feature channels to be used in the ConvL, and f is the activation function. m, i and j are the serial numbers of convolutional layer, input and output feature channels, respectively.

ReLU activation function is used, which directly sets the negative features to zeros. The major benefit of using ReLU is that its derivative value is either 1 or 0, and is helpful to mitigate the risk of vanishing and exploding gradients. The ReLU activation function is expressed by

$$x \leftarrow \max(x, 0) \tag{6}$$

BN [39] is a special adaptive normalization module that is inserted inside the deep neural networks, in order to reduce internal covariate shift and thereby accelerate the training process. In other words, the distributions of features in different batches will become more stable and adjustable with the use of BNs. The formula of a BN module are expressed by

$$\mu = \frac{1}{N_{\text{batch}}} \sum_{n=1}^{N_{\text{batch}}} x_n \tag{7}$$

$$\sigma^2 = \frac{1}{N_{\text{batch}}} \sum_{n=1}^{N_{\text{batch}}} (x_n - \mu)^2 \tag{8}$$

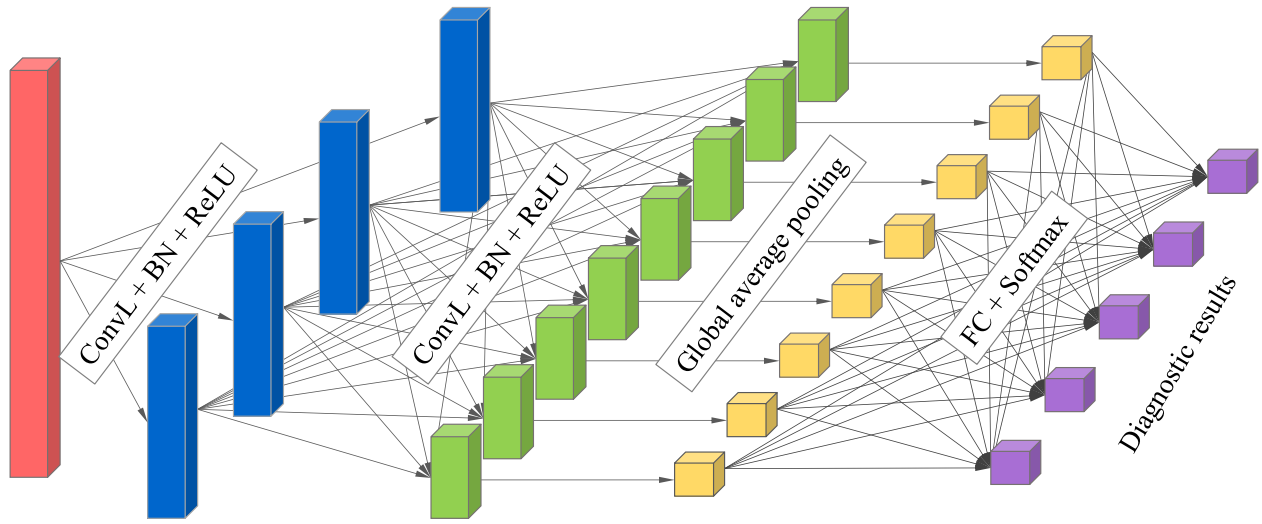


Fig. 3. Architecture of a typical ConvNet.

$$\widehat{x}_n = \frac{x_n - \mu}{\sqrt{\epsilon + \sigma^2}} \tag{9}$$

$$y_n = \beta + \gamma \widehat{x}_n \tag{10}$$

where N_{batch} is the number of samples in a batch, ϵ is a near-zero positive value, β and γ are two trainable parameters that shifts and rescales the feature distributions.

Global average pooling constructs an average from each feature channel. The number of neurons and weights is greatly reduced in the final FC layer, which further reduces the effect of overfitting and eases the difficulty of optimization.

At the end of the ConvNet, the softmax cross-entropy loss is employed as the objective function to be minimized. The softmax function, which converts the features into the range of $(0,1)$, is expressed by

$$p_i = \frac{e^{y_i}}{\sum_{j=1}^{N_{\text{class}}} e^{y_j}} \tag{11}$$

where y_i is the i th input feature, N_{class} is the number of known classes (*i. e.*, health conditions), and p_i is the estimated probability that the sample belonging to the i th class. Then, the cross-entropy loss L, which measures the distance between the real probability t_j and the estimated probability p_j, is calculated by

$$L = -\sum_{j=1}^{N_{\text{class}}} t_j \log(p_j) \tag{12}$$

Finally, the gradient back-propagation algorithm is applied to optimize the stochastically initialized weights and zero-initialized biases in the ConvNet through a number of epochs. The optimized ConvNet can be applicable to the imbalanced fault diagnosis tasks.

2.3. Procedure of imbalanced fault diagnosis

The overall procedure of the developed imbalanced fault diagnosis approach is shown in Fig. 4. Firstly, a training dataset is prepared, which is composed of a sufficient number of healthy samples and a very tiny number of faulty samples. Secondly, the number of faulty samples is enlarged using the developed wavelet packet distortion, in order to make every faulty class has the same number of samples with the healthy class, so that these classes are balanced. Thirdly, train the ConvNet model for 1 epoch using the balanced training dataset. Fourthly, if the training accuracy reaches 100% or this is the 200th epoch, the training is terminated; otherwise, the faulty samples are augmented again, and the ConvNet model is trained for 1 epoch again; the purpose of this arrangement is to prevent the overfitting issue. Finally, the trained ConvNet model is applied for fault diagnosis of mechanical systems.

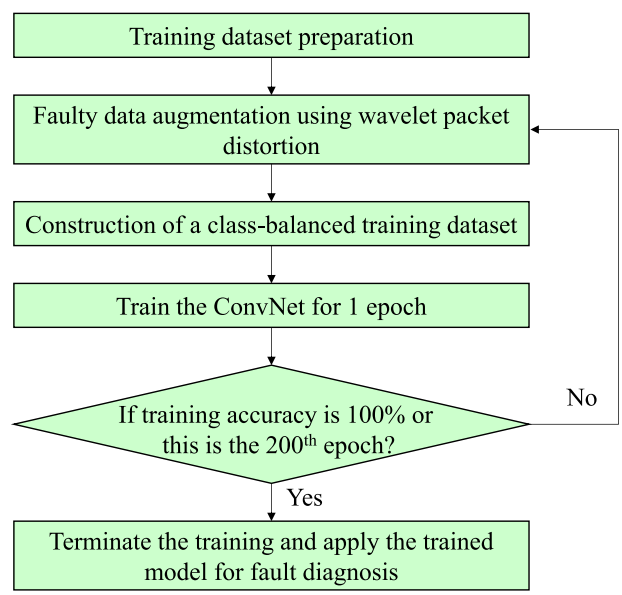


Fig. 4. Overall procedure of the imbalanced fault diagnosis approach.

3. Experimental validation

The developed imbalanced fault diagnosis approach for mechanical systems was implemented using TensorFlow 1.0, and the experiments were conducted on an ordinary personal computer with Intel i7-10700K CPU and NVIDIA RTX3060 GPU. The experimental data, setups, and results are introduced as below.

3.1. Experimental data

As summarized in Table 1, vibration signals under 6 health conditions (including 1 healthy condition and 5 faulty conditions) were collected from an on-ground simulation test rig for civil aero hydraulic pumps. Each sample is a particular length of vibration signal with 128 data points. In order to simulate the class imbalance, the healthy class

Table 1
Descriptions on the training and testing data.

Labels	Descriptions	Amount of training samples	Amount of testing samples
H	No observable faults	400	80
F1	Wear-out failure of oil distribution pan	5	475
F2	Wear-out failure of cylinder blocks	5	475
F3	Eroding failure with 0.3 Mpa lower entry pressure	5	475
F4	Distortion failure with 20 Mpa exit pressure	5	475
F5	Fatigued failure with a chink of 0.8 mm	5	475

has 400 training samples, while each faulty class only has 5 training samples; the imbalance ratio is 80:1. The other samples in the obtainable dataset are put into the testing dataset. The healthy class has 80 testing samples, but each faulty class contains 475 testing samples.

3.2. Hyperparameter configurations

The hyperparameters can be classified into two groups, including the wavelet-related hyperparameters and ConvNet-related hyperparameters. First, for wavelet-related hyperparameters, the decomposition level is 3, and "DB4" wavelet function is used due to its wide applications in fault diagnosis tasks [40,41]. Second, for ConvNet-related hyperparameters, the number of convolutional layers is 2; the numbers of convolutional kernels in the two convolutional layers are 8 and 16, respectively; the length of convolutional kernels is 3; the learning rate is set to be 0.1 in the first epoch, and multiplied with 0.96 in each of the next epoch; the momentum coefficient is 0.9; the weight decay in L2 regularization is 0.0001. Finally, the training process is terminated at the 200th epoch or when the training accuracy becomes 100%.

3.3. Experimental comparison with traditional methods

The developed method is not only compared with the ConvNets without any manipulations on the training dataset, but also compared with the down sampler (DS) + ConvNet and over sampler (OS) + ConvNet. Here, the range of distortion coefficients is [0.8, 1.2) in the developed method; DS means that 5 samples are stochastically selected from the healthy class, and used as a part of the training dataset; OS means that all faulty samples are simply copied for a number of times to achieve a balance between the classes. Each experiment has been repeated 10 times with different samples as the training dataset and the other samples as the testing dataset.

The F1, precision, and recall of experimental results are displayed in Figs. 5–7. As we can see, the developed method yielded the highest performance in 7 among the 10 tests. In the 2nd, 5th, and 8th tests, the developed method is inferior to the OS + ConvNet. The major reason is

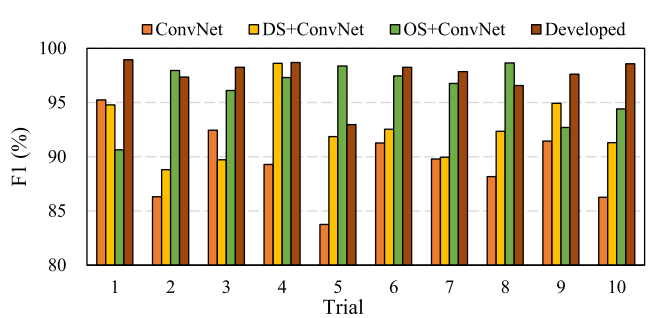


Fig. 5. F1 scores of the four mentioned methods in 10 trials.

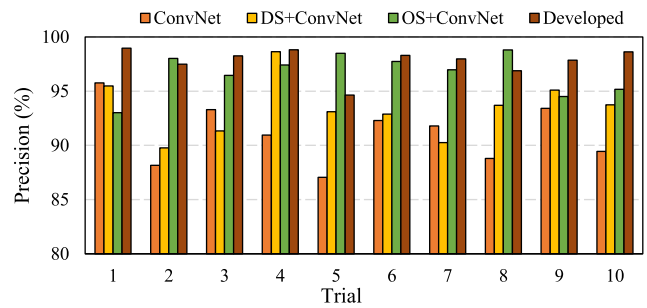


Fig. 6. Precisions of the four mentioned methods in 10 trials.

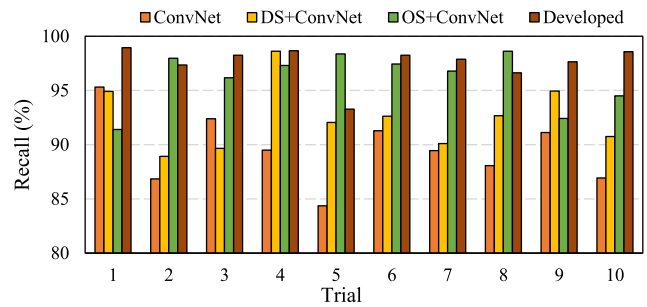


Fig. 7. Recalls of the four mentioned methods in 10 trials.

that there is a high randomness in the tests; there are only 5 samples in each faulty class. If the 5 samples cannot fully represent the whole class, it will not be probable for the methods to guarantee a good performance.

The averages of evaluation metrics in Figs. 5–7 are summarized in Table 2. Obviously, the average performance of the developed method is the best among the four methods. It achieved 1.47%, 1.12%, and 1.44% improvements compared to the OS + ConvNet in terms of the three average metrics, which validated that the wavelet packet distortion can not only improve the diversity of training dataset, but also improve the accuracies of imbalanced fault diagnosis of aero hydraulic pumps.

The stabilities of these methods are further compared using a box plot in Fig. 8. It can be seen that most of the results of the developed method distribute in smaller ranges compared to the other methods. Here, the unstable performance mainly has two reasons, *i.e.* (1) overfitting and (2) characteristics of training fault samples. On one hand, the developed wavelet packet distortion creates various training samples and reduces overfitting, so that the developed method is more stable than the other methods. On the other hand, sometimes, the randomly selected training fault samples happen to be very different from the testing fault samples, so that the performance of all these methods cannot be stable in this situation; this is also a major reason for the outlier (in Fig. 8) of the developed method.

The changing trends of training and testing losses during the training phases are depicted in Fig. 9. It is observable that the training of the first two methods (*i.e.*, ConvNet and DS + ConvNet) was terminated after more than 30 epochs, while the training processes of the other two

Table 2
A summary of average results in Figs. 5–7 (%).

Average metric	Method			
	ConvNet	DS + ConvNet	OS + ConvNet	Developed
F1	89.40 ± 3.40	92.48 ± 2.95	96.03 ± 2.65	**97.50 ± 1.74**
Precision	91.10 ± 2.73	93.40 ± 2.63	96.66 ± 1.88	**97.78 ± 1.27**
Recall	89.52 ± 3.16	92.53 ± 2.97	96.10 ± 2.51	**97.54 ± 1.65**

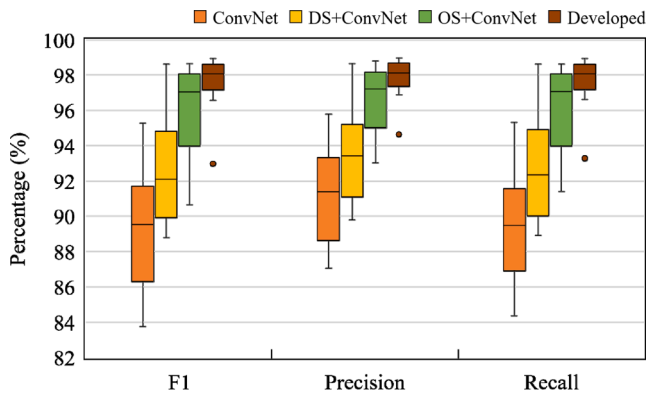


Fig. 8. A box plot of the results in Figs. 5–7.

methods (*i.e.*, OS + ConvNet and the developed method) were terminated at the 5th epoch. The reason is that there are only $400 + 5 \times 5 = 425$ and $5 \times 6 = 30$ training samples in the ConvNet and DS + ConvNet, respectively, but there are $400 \times 6 = 2400$ training samples in the OS + ConvNet and the developed method. Besides, an interesting thing is that the training losses of the OS + ConvNet and the developed method are relatively low (around 0.1) at the first epoch. The reason is that the losses reduced to the level around 0.1 after just 1 epoch. The training dataset is not difficult for ConvNets to fit. Most importantly, the testing loss of our developed method is around 0.05 and obviously lower than the other three methods, which validates that the developed method has a relatively good generalizability on the testing dataset.

Besides, the developed method does not cost much time. On one hand, wavelet packet decomposition and reconstruction are computational efficient. On the other hand, a small training dataset is used in this paper. To be specific, the training time per epoch of the developed method is about 8 s (including the time for wavelet packet distortion and deep model optimization), so that the total training time in Fig. 9(d) should be around $8\ \text{s} \times 5$ epochs $= 40$ s. Although sometimes the developed method needs a few more epochs, it will not cost much longer training time.

The changing trends of the F1 scores are depicted in Fig. 10. It is observable that the final testing F1 value of the developed method is higher than the first three methods, which corresponds to the highest performance. Besides, the training and testing F1 scores of the developed method are relatively low at the 2nd epoch, which is probably caused by the fact that some augmented faulty samples may become too different from the real faulty samples, and they misguide the optimization of related parameters.

The learned high-dimensional features are converted into two dimensional spaces using t-distributed stochastic neighbor embedding [42]. The distributions of testing samples in the learned feature spaces are depicted in Fig. 11. It is observable that the F1 and F2 classes highly overlapped with each other in the ConvNet, which indicates that the two classes are not so distinguishable. Then, the misclassifications in DS + ConvNet and OS + ConvNet are observably more than the developed method, which validate that the developed method has a high capability in learning discriminant features.

To have a clear observation on the diagnostic performance, the confusion matrices of the four methods are provided in Fig. 12. It is observable that the accuracies on the F1 class are 71%, 86%, 80%, and 94%, respectively, when using the four methods. Further, the class-wise accuracies of the developed method are mostly higher than the other three methods. Therefore, the performance of our developed method is more reliable for most of the health conditions in these experiments.

3.4. Performance when using different ranges of distortion coefficients

The distortion coefficient has a direct impact on the augmented samples. If the distortion coefficient is far away from 1.0, the augmented sample will be more dissimilar from the corresponding original sample. On the contrary, if the distortion coefficient is close to 1.0, the augment sample will be close to the original sample. Then, if a larger range of distortion coefficients is adopted, there will be more variety in the augmented dataset, so that a relatively larger ConvNet is required.

In this subsection, 5 different ranges of distortion coefficients are tested, including $[0.5, 1.5)$, $[0.6, 1.4)$, $[0.7, 1.3)$, $[0.8, 1.2)$, and $[0.9, 1.1)$.

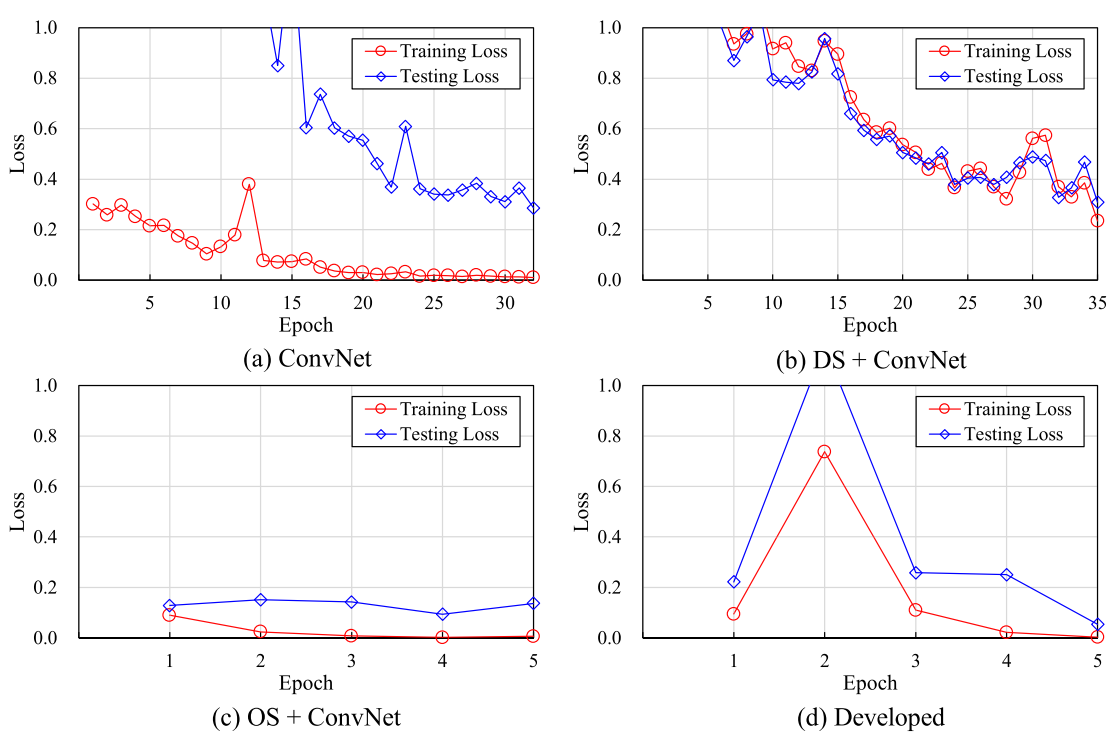


Fig. 9. The changing trends of losses during the training phases.

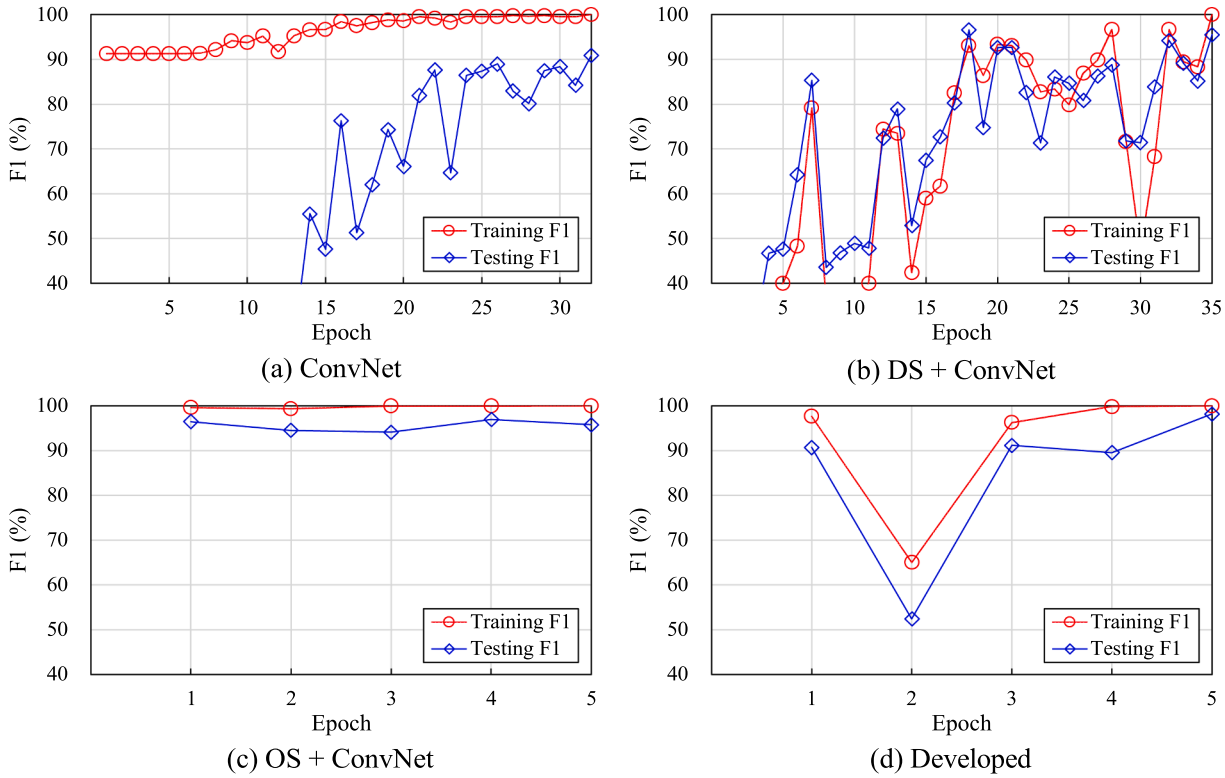


Fig. 10. The changing trends of F1 scores during the training phases.

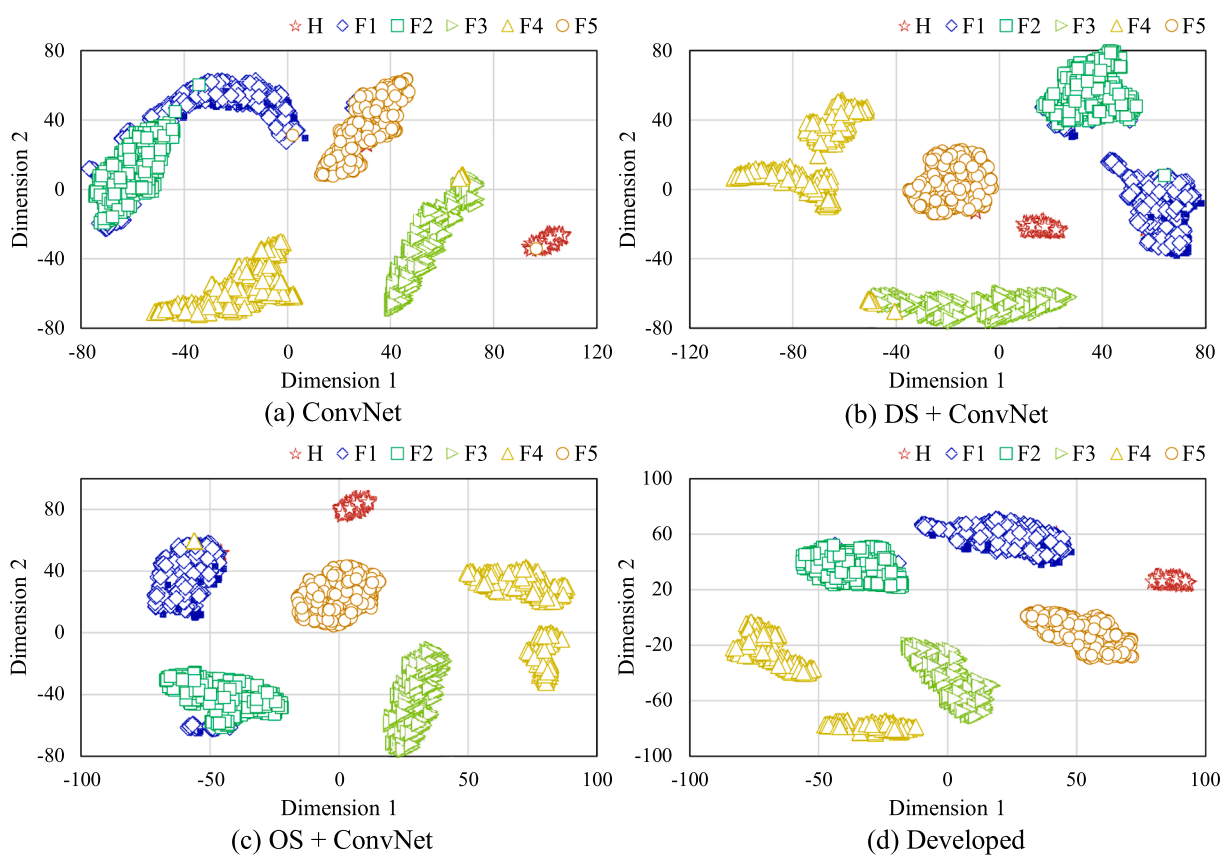


Fig. 11. Distributions of testing samples in the two dimensional feature spaces.

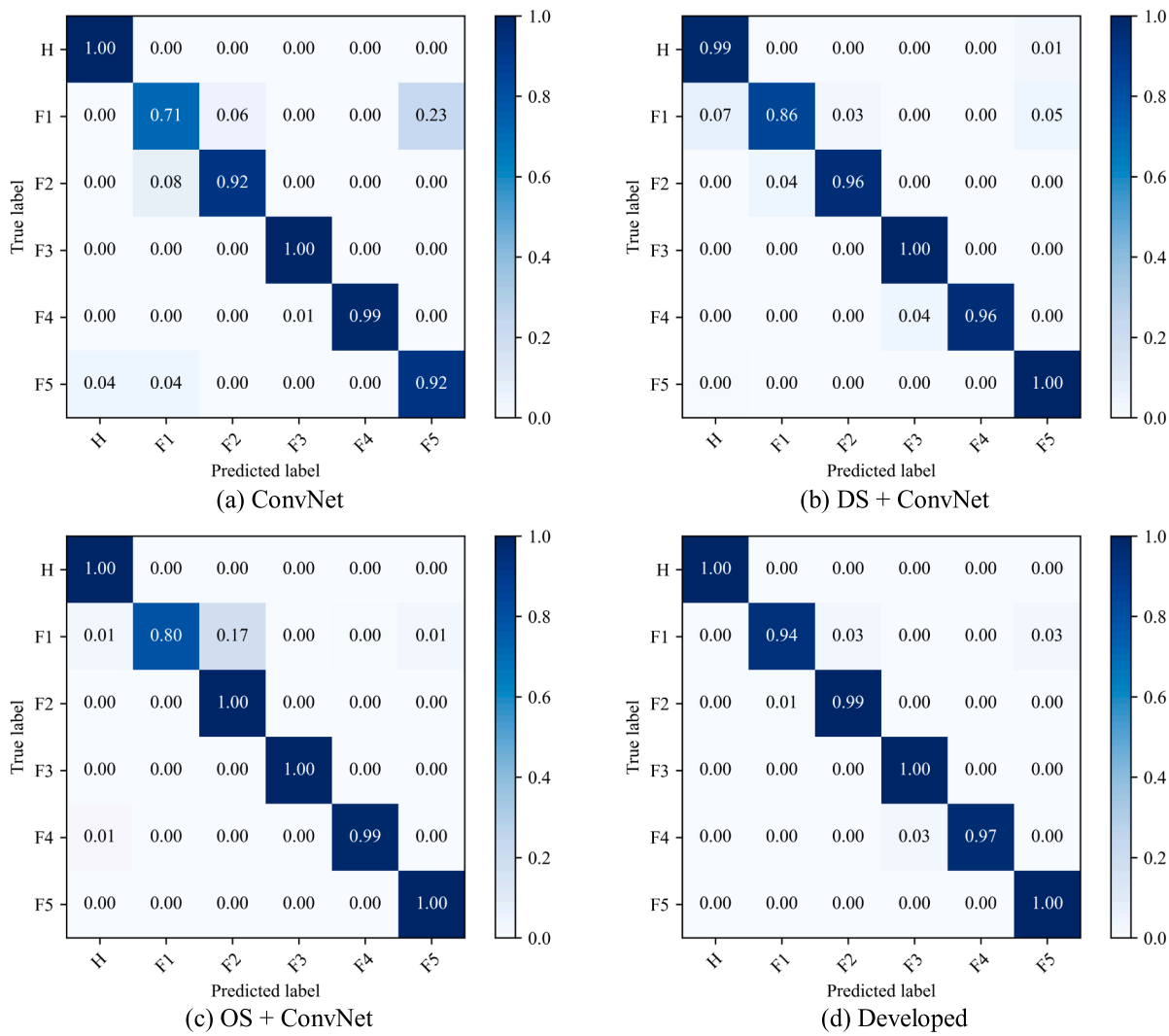


Fig. 12. Confusion matrices of the considered methods on a testing dataset.

Table 3
F1 scores of the developed method when using different ranges of distortion coefficients (%).

Test	Range of distortion coefficients					
	[0.5,1.5)	[0.6, 1.4)	[0.7,1.3)	[0.8, 1.2)	[0.9, 1.1)	None
Trial 1	98.32	97.10	**99.14**	98.53	95.12	98.25
Trial 2	97.66	97.81	97.82	**98.25**	97.01	96.55
Trial 3	97.24	**98.86**	98.82	97.02	97.40	94.52
Trial 4	99.51	99.06	98.94	**99.59**	98.85	98.16
Trial 5	**99.43**	99.27	98.28	98.12	98.98	98.70
Trial 6	98.77	98.65	99.55	**99.92**	99.39	97.19
Trial 7	97.23	96.37	96.59	95.29	**97.51**	97.20
Trial 8	**99.59**	98.77	98.86	99.27	97.91	97.56
Trial 9	**97.58**	97.08	97.00	96.00	96.60	97.26
Trial 10	96.11	96.22	96.44	**96.68**	95.97	95.15
Average ± STD	**98.14 ± 1.17**	97.92 ± 1.15	**98.14 ± 1.12**	97.87 ± 1.57	97.47 ± 1.37	97.05 ± 1.33

Table 4
Precisions of the developed method when using different ranges of distortion coefficients (%).

Test	Range of distortion coefficients					
	[0.5,1.5)	[0.6, 1.4)	[0.7,1.3)	[0.8, 1.2)	[0.9, 1.1)	None
Trial 1	98.37	97.31	**99.15**	98.55	95.65	98.44
Trial 2	97.77	97.94	97.92	**98.35**	97.25	96.77
Trial 3	97.37	**98.88**	98.83	97.12	97.42	95.08
Trial 4	99.52	99.07	98.96	**99.60**	98.88	98.29
Trial 5	**99.43**	99.27	98.35	98.24	98.99	98.72
Trial 6	98.82	98.69	99.56	**99.92**	99.39	97.45
Trial 7	97.42	96.68	96.87	95.79	**97.74**	97.24
Trial 8	**99.60**	98.79	98.88	99.27	98.05	97.70
Trial 9	**97.80**	97.40	97.27	96.56	97.07	97.49
Trial 10	96.54	96.61	96.82	**97.04**	96.50	95.83
Average ± STD	**98.26 ± 1.05**	98.06 ± 1.00	**98.26 ± 0.99**	98.04 ± 1.37	97.69 ± 1.17	97.30 ± 1.15

The corresponding results have been summarized in Tables 3–5, in which "none" means that the distortion coefficient always equals 1.0, *i.e.*, the faulty samples are simple copied without any distortions. Although $[0.5, 1.5)$ and $[0.7, 1.3)$ correspond to the highest average performance, they only achieve the highest performance for 3 and 1 times, respectively. The other 3 ranges (*i.e.*, $[0.6, 1.4)$, $[0.8, 1.2)$, and $[0.9, 1.1)$) have 1, 4, and 1 times to yield the best performance, respectively. Therefore, it is not appropriate to determine one range to be the optimum. Further, the choice of range should be considered alongside with the size of ConvNet. Aiming at this problem, a validation dataset can be

Table 5
Recalls of the developed method when using different ranges of distortion coefficients (%).

Test	Range of distortion coefficients					
	[0.5,1.5)	[0.6, 1.4)	[0.7,1.3)	[0.8, 1.2)	[0.9, 1.1)	None
Trial 1	98.33	97.11	**99.14**	98.53	95.15	98.21
Trial 2	97.68	97.84	97.84	**98.25**	97.03	96.50
Trial 3	97.27	**98.86**	98.82	97.03	97.39	94.34
Trial 4	99.51	99.06	98.94	**99.59**	98.86	98.17
Trial 5	**99.43**	99.27	98.29	98.13	98.98	98.70
Trial 6	98.78	98.66	99.55	**99.92**	99.39	97.23
Trial 7	97.27	96.46	96.62	95.40	**97.52**	97.23
Trial 8	**99.59**	98.78	98.86	99.27	97.92	97.56
Trial 9	**97.60**	97.11	97.07	96.05	96.62	97.31
Trial 10	96.21	96.33	96.54	**96.74**	96.05	95.32
Average ± STD	**98.17 ± 1.15**	97.95 ± 1.12	**98.17 ± 1.09**	97.89 ± 1.53	97.49 ± 1.35	97.05 ± 1.35

prepared and used to select an appropriate distortion range for the testing dataset in the real applications.

4. Conclusions

This paper develops a new imbalanced fault diagnosis method for mechanical systems, which uses wavelet packet distortion to augment faulty samples with the objective of achieving the balance between the healthy and faulty classes, and applies a typical deep learning method, *i.e.*, ConvNet, to diagnose the faults through classification.

The developed method yielded 8.10%, 5.02%, and 1.47% improvements compared to the ConvNet, DS + ConvNet, OS + ConvNet, respectively, in terms of average F1. These improvements are mainly owing to the use of wavelet packet distortion as a data augmentation method to achieve a balance among the classes. With an augmented training dataset, the subsequent ConvNet does not encounter the class imbalance and overfitting issues, and achieves higher recognition accuracies.

Besides, this paper investigated the impact of value ranges for stochastically selecting distortion coefficients, and experimental results indicated that the ConvNets with wavelet packet distortions mostly yielded higher performance than the ConvNets without wavelet packet distortions, which validated that the developed method has a quite stable performance. It is also suggested to prepare a validation dataset to determine the value range in real applications.

Finally, the idea of wavelet packet distortion can be applicable to imbalanced (or few-shot) fault diagnosis of other mechanical systems, such as wind power generators, gas turbines and civil aviation engines. It is also usable for data augmentation of other types of time-series signals alongside with other deep learning methods, and it has a latent capacity to improve pattern recognition accuracies in the other research fields. Besides, it is notable that this paper directly uses the “DB4” wavelet function, which still does not have a solid theoretical justification at this moment. That is to say, some other wavelet functions may be more suitable than “DB4” in this task. Future research will be conducted on how to select or design a proper wavelet function with solid justifications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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